Durethan® ACF30XH2.0EF

PA66-(GF+CF)30 — Envalior

Injection Molding, 30% Glass Fiber/Carbon Fiber Reinforced, Heat Stabilized, Electrical Conductive, High Flow

ISO 1043 PA66-(GF+CF)30

Rheological properties	dry / cond	Unit	Test Standard
ISO Data			
Molding shrinkage, parallel	0.3 / *	%	ISO 294-4, 2577
Molding shrinkage, normal	0.8 / *	%	ISO 294-4, 2577

Mechanical Properties	dry / cond	Unit	Test Standard
ISO Data			
Tensile Modulus	12000 / 7400	MPa	ISO 527
Stress at Break	170 / 110	MPa	ISO 527
Strain at Break	2.5 / 4.5	%	ISO 527
Impact Strength (Charpy), +23°C	60 / 65	kJ/m²	ISO 179/1eU
Impact Strength (Charpy), -30°C	60 / 55	kJ/m²	ISO 179/1eU
Notched Impact Strength (Charpy), +23°C	- / 15	kJ/m²	ISO 179/1eA
Puncture - maximum force, +23°C	840 / -	N	ISO 6603-2
Puncture - maximum force, -30°C	700 / -	N	ISO 6603-2
Puncture energy, +23°C	3.1 / -	J	ISO 6603-2
Puncture energy, -30°C	2.3 / -	J	ISO 6603-2

Thermal Properties	dry / cond	Unit	Test Standard
ISO Data			
Melting Temperature (10°C/min)	261 / *	°C	ISO 11357-1/-3
Temp. of deflection under load (1.80 MPa)	240 / *	°C	ISO 75-1/-2
Temp. of deflection under load (0.45 MPa)	250 / *	°C	ISO 75-1/-2
Coeff. of Linear Therm. Expansion, parallel	10 / *	E-6/K	ISO 11359-1/-2
Coeff. of Linear Therm. Expansion, normal	100 / *	E-6/K	ISO 11359-1/-2

Electrical Properties	dry / cond	Unit	Test Standard
ISO Data			
Volume Resistivity	1000 / -	Ohm*m	IEC 62631-3-1

Other Properties	dry / cond	Unit	Test Standard
ISO Data			
Water Absorption	5 / *	%	Sim. to ISO 62
Humidity absorption	1.7 / *	%	Sim. to ISO 62
Density	1300 / -	kg/m³	ISO 1183

Test specimen production	Value	Unit	Test Standard
ISO Data			
Injection Molding, melt temperature	290	°C	ISO 294
Injection Molding, mold temperature	80	°C	ISO 294

Processing Recommendation Injection Molding	Value	Unit	Test Standard
Pre-drying - Temperature	80	°C	-
Pre-drying - Time	2 - 6	h	-
Processing humidity	≤0.07	%	-
Melt temperature	280 - 300	°C	-
Mold temperature	80 - 120	°C	-

Characteristics

Processing

Injection Molding

Special Characteristics

Electrically Conductive, Heat aging stabilized

Delivery form

Pellets

Injection Molding

PREPROCESSING
Residual moisture content: 0.03 - 0.07%
Drying temperature dry air dryer: 80 °C

Drying time dry air dryer 2 - 6 h

PROCESSING
Melt temperature (Tmin - Tmax): 280 - 300 °C

Mold temperature: 80 - 120 °C

Disclaimer

Liability Exclusion
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